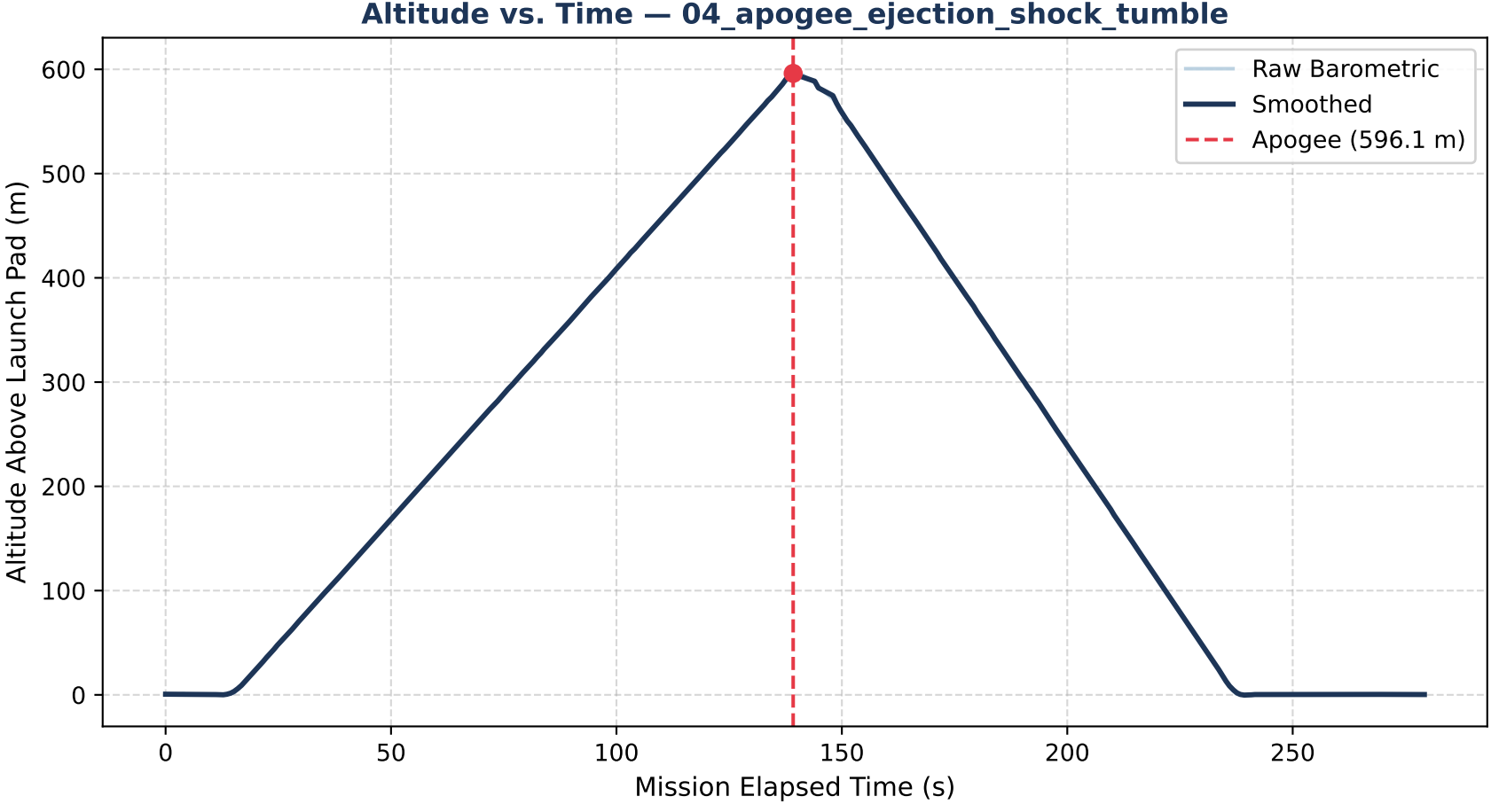
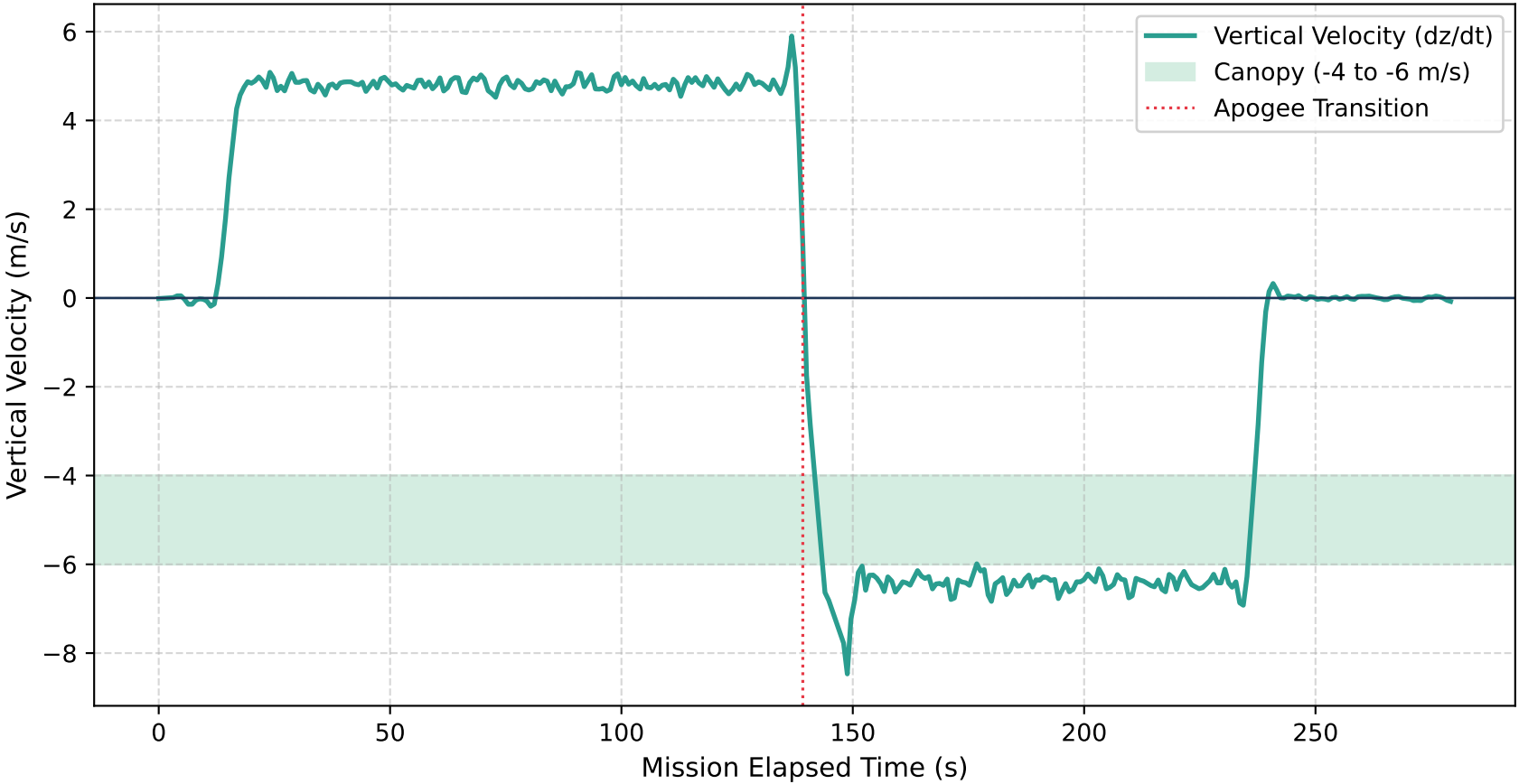
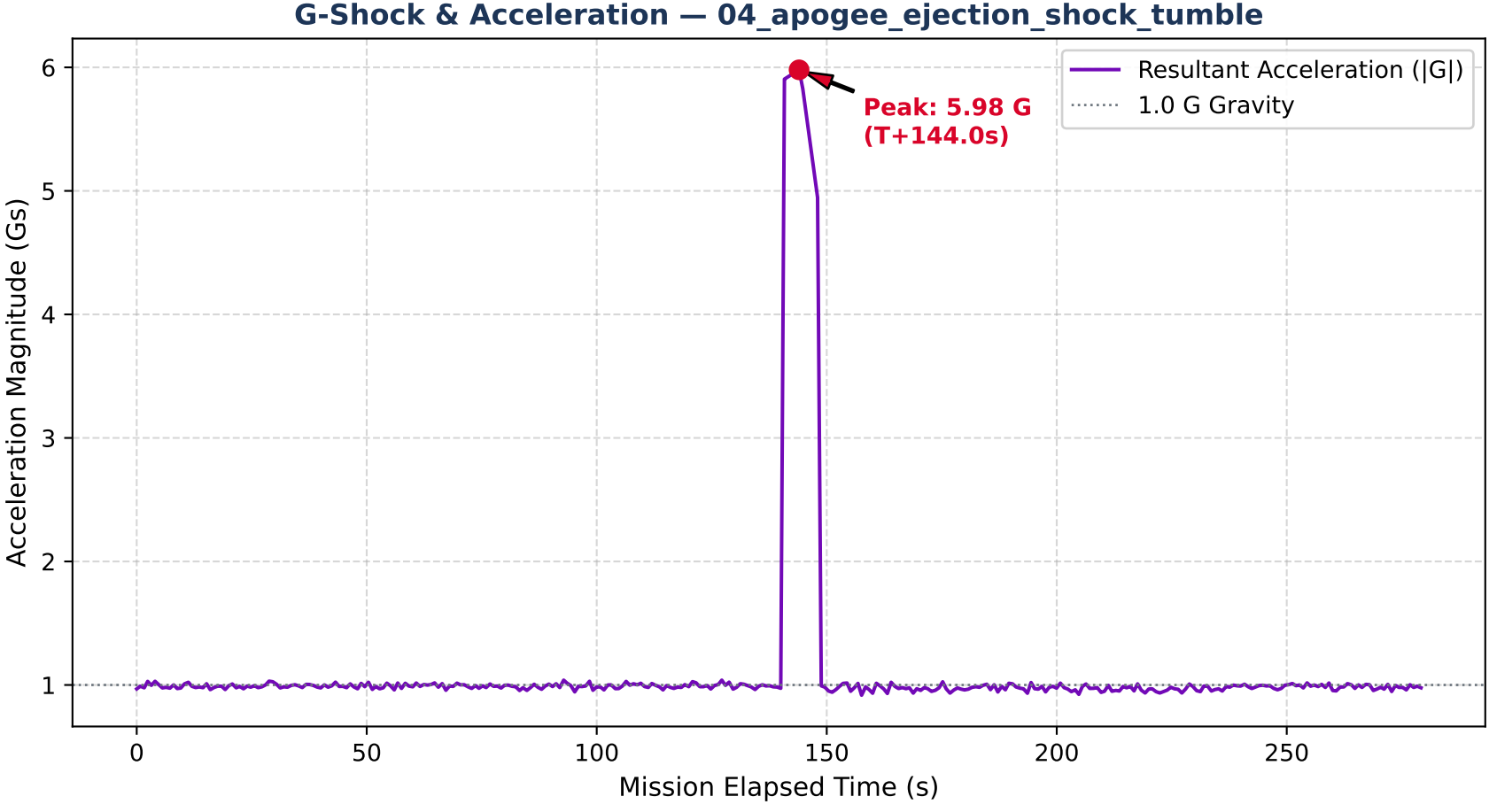
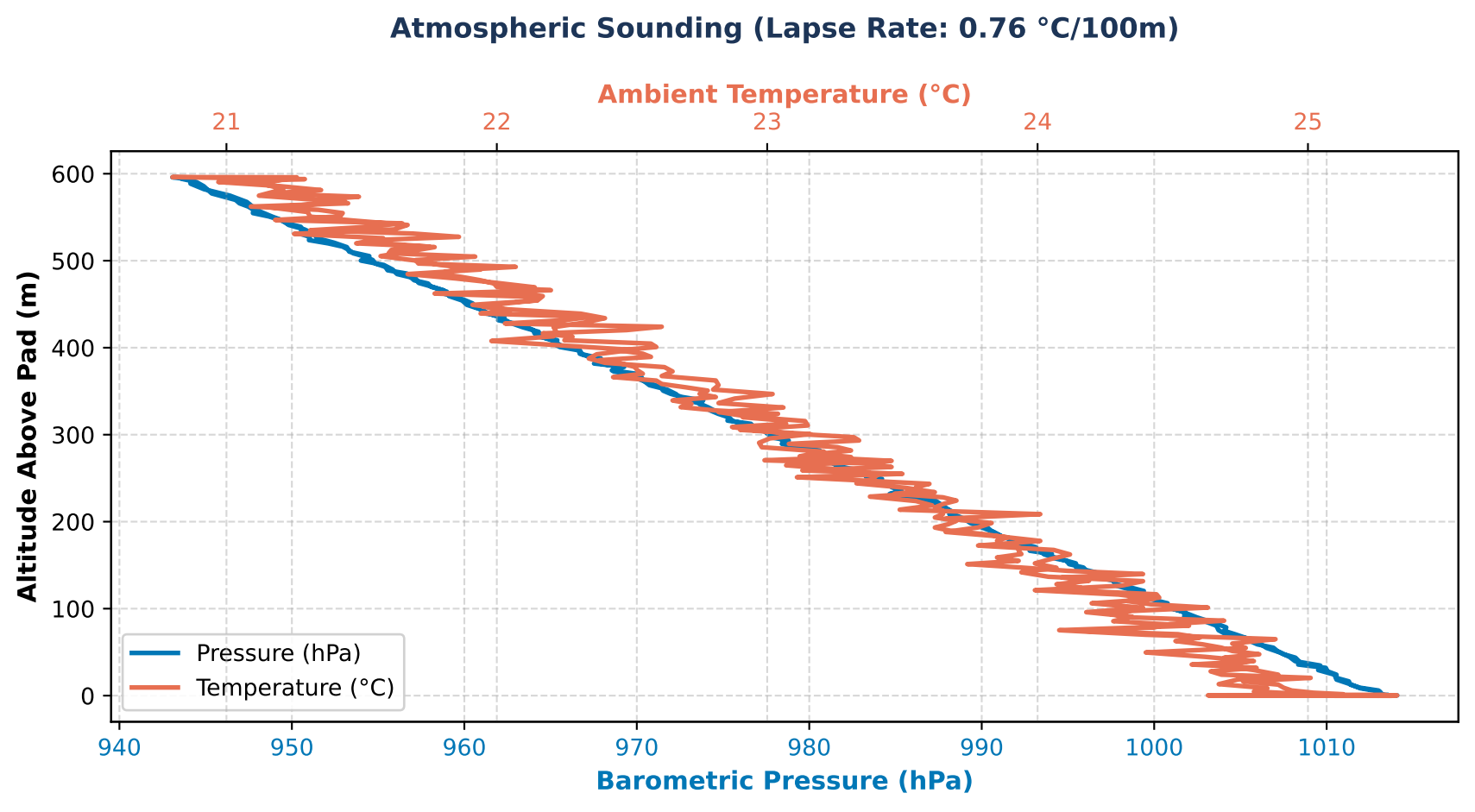
Cognitive CanSat - Mission Summary: 04_apogee_ejection_shock_tumble

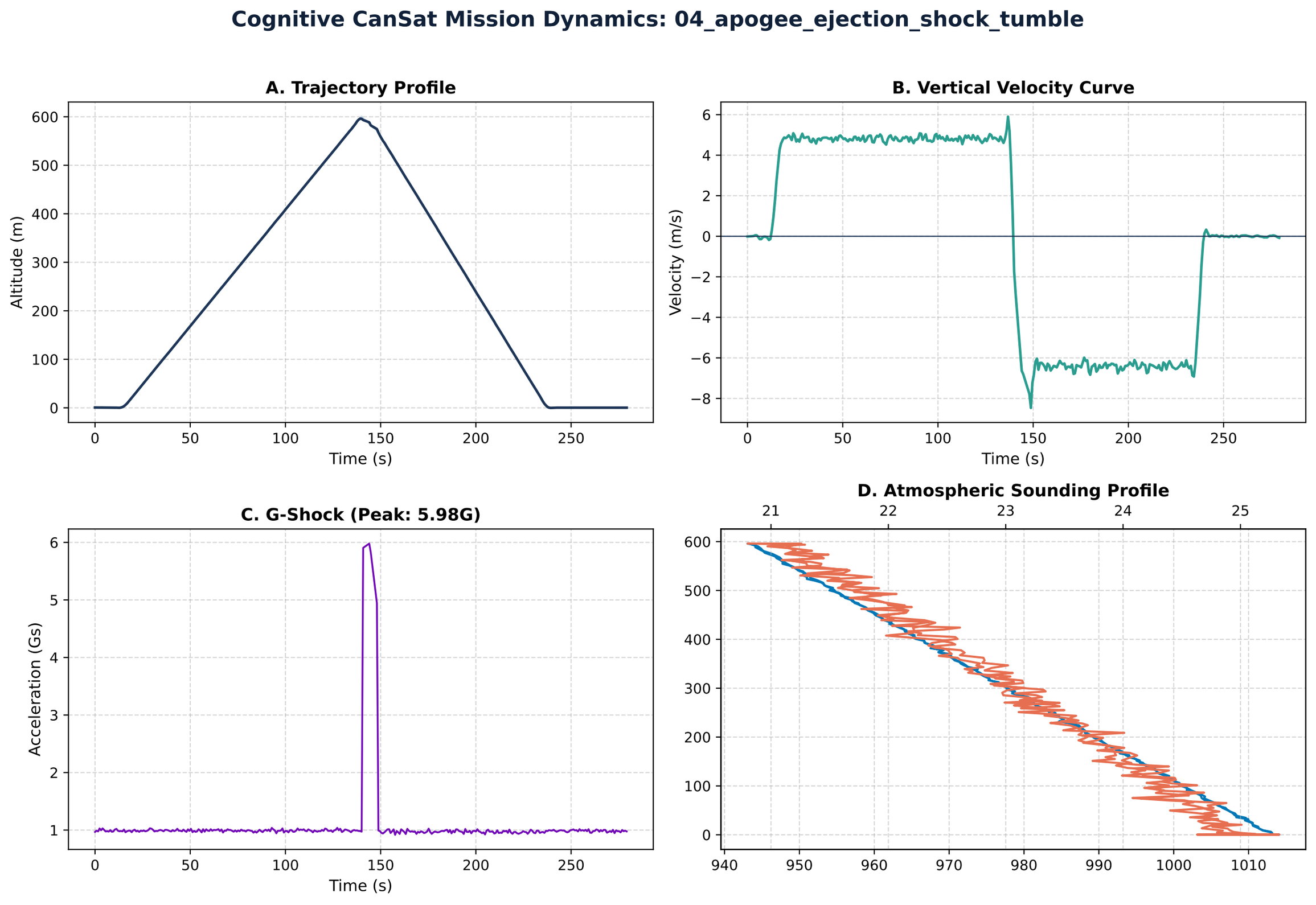
Flight Dynamics Parameter	Calibrated Value
Mission Name	04_apogee_ejection_shock_tumble
Apogee (m)	596.1
Time to Apogee (s)	139.2
Max Ascent Velocity (m/s)	5.9
Max Descent Velocity (m/s)	-8.47
Max G-Shock (Gs)	5.98
Touchdown Shock (Gs)	1.02
Mission Duration (s)	279.2
Min Temperature (°C)	20.8
Max Temperature (°C)	25.33
Min Pressure (hPa)	943.07
Max Pressure (hPa)	1014.09
Lapse Rate (°C/100m)	0.76
Telemetry Frames	344
Landing Lat	22.575554
Landing Lon	88.368961


Altitude vs. Time — 04_apogee_ejection_shock_tumble
Raw Barometric
Smoothed
Apogee (596.1 m)
Altitude Above Launch Pad (m)
Mission Elapsed Time (s)
0
100
200
300
400
500
600
0
50
100
150
200
250


Vertical Velocity Profile — 04_apogee_ejection_shock_tumble
Vertical Velocity (dz/dt)
Canopy (-4 to -6 m/s)
Apogee Transition
Vertical Velocity (m/s)
6
4
2
0
−2
−4
−6
−8
0
50
100
150
200
250
Mission Elapsed Time (s)


G-Shock & Acceleration — 04_apogee_ejection_shock_tumble
Resultant Acceleration (|G|)
1.0 G Gravity
Peak: 5.98 G
(T+144.0s)
Acceleration Magnitude (Gs)
Mission Elapsed Time (s)
1
2
3
4
5
6
0
50
100
150
200
250


Atmospheric Sounding (Lapse Rate: 0.76 °C/100m)
Ambient Temperature (°C)
21
22
23
24
25
Altitude Above Pad (m)
600
500
400
300
200
100
0
Pressure (hPa)
Temperature (°C)
940
950
960
970
980
990
1000
1010
Barometric Pressure (hPa)


Cognitive CanSat Mission Dynamics: 04_apogee_ejection_shock_tumble
A. Trajectory Profile
Altitude (m)
Time (s)
B. Vertical Velocity Curve
Velocity (m/s)
Time (s)
C. G-Shock (Peak: 5.98G)
Acceleration (Gs)
Time (s)
D. Atmospheric Sounding Profile